

New Procedure for Quantitative Elemental Analysis by Laser-Induced Plasma Spectroscopy

A. CIUCCI, M. CORSI, V. PALLESCHI, * S. RASTELLI, A. SALVETTI, and E. TOGNONI

Istituto di Fisica Atomica e Molecolare del CNR, Via del Giardino, 7-56127 Pisa, Italy

A new procedure, based on the laser-induced plasma spectroscopy (LIPS) technique, is proposed for calibration-free quantitative elemental analysis of materials. The method here presented, based on an algorithm developed and patented by IFAM-CNR, allows the matrix effects to be overcome, yielding precise and accurate quantitative results on elemental composition of materials without use of calibration curves. Some applications of the method are illustrated, for quantitative analysis of the composition of metallic alloys and quantitative determination of the composition of the atmosphere.

Index Headings: Lasers; Plasma. Laser-induced plasma spectroscopy (LIPS); Quantitative analysis.

INTRODUCTION

The first applications of laser-induced plasma spectroscopy (LIPS) for diagnostic purposes were proposed in the early 1980s.¹⁻⁶ Since then, many LIPS-based diagnostic systems have been developed for research and industrial applications,⁷⁻²⁹ taking advantage of the intrinsic characteristics of this technique, such as quickness, sensitivity, and feasibility of *in situ* measurements. In particular, LIPS has been extensively used as a reliable analysis technique for trace element detection in the atmosphere^{8,9} as well as in liquid effluents^{11,12} or in metallic alloys.^{14,15}

In last years, several works have studied the possibility of LIPS applications to quantitative elemental analysis of materials, by comparing the LIPS signal obtained for a given element to a suitable calibration curve.^{17-20,24-29} However, the LIPS signal emitted by the same element depends on the matrix in which it is embedded;^{24,29} this "matrix effect" leads to the necessity of many calibration curves—in principle, one for any substrate composition. In the case of samples of unknown composition, this requirement would lead to the paradoxical need for an exact knowledge of the sample composition in order to build the proper calibration curve required for the measurement. This problem has been studied in recent papers by Xu et al.³⁰ and Chaléard et al.,³¹ where two different approaches have been suggested in order to alleviate the matrix effect in LIPS analysis.

However, at the present time, no exhaustive approach has been suggested that is able to completely overcome this problem. In practical situations, the use of empirical calibration curves might be useful for measurements of the concentration of a single species in a well-defined matrix;^{8,12} however, this procedure would be inappropriate in more complex cases, such as multielemental analysis in *a priori* unknown environments.

Hopefully, the basic physics of the laser-induced plasma process will allow us to establish a measurement protocol that is not affected by the above-mentioned effects. Assuming that some simple conditions are verified, a quantitative analysis of the atomic species constituting the sample is possible, without calibration curves. This calibration-free procedure, applicable to solids, liquids, particulate, and gaseous samples, will be described in the next section.

DESCRIPTION OF THE PROCEDURE

Working Hypothesis. In order to develop a calibration-free measurement protocol we should make some reasonable assumptions concerning both the experimental method of operation and the theoretical representation of the plasma conditions.

First, we will assume that the plasma composition is representative of the actual material composition prior to the ablation (this is the basic assumption for the application of the LIPS technique, which is well realized in the range of the laser energies used in LIPS experiments).³² Moreover, we assume that in the actual temporal and spatial observation window, the plasma is in local thermal equilibrium (LTE) conditions and the radiation source is optically thin (this hypothesis is in general fulfilled in LIPS experiments, where the main ionization process is produced through impact excitation by thermal electrons).³³

Although these requirements might seem exacting, experimental results obtained in our laboratory showed that the population of the excited atomic levels (with typical energy between 3 eV and 8 eV) obeys the Boltzmann distribution. This is a good *a posteriori* demonstration that the plasma produced is close to LTE conditions. In normal LIPS operating conditions, the assumption of optically thin plasma is fulfilled for trace elements, while in a few cases it may become critical for intense resonance lines of the primary components. The self-absorption effect can be theoretically modeled, and a forthcoming paper will be published by the authors on this topic.¹⁵ At the moment, our analysis will consider only the spectral lines that are not self-absorbed.

In our experimental conditions, only lines generated by neutral species and single ionized species can be detected. In order to distinguish between the different ionization states of the same atomic element, in the following discussion we will refer to them as "species", i.e., atomic elements with a determined electric charge (0 or +1). For instance, Mg(I) and Mg(II) are two distinct species of the same atomic element, with charges 0 and 1, respectively.

In an LTE approximation, the line integral intensity

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* Author to whom correspondence should be sent.

corresponding to the transition between two levels E_k and E_i of an atomic species s can be expressed as

$$I_{ki}^{\lambda} = N_s A_{ki} \frac{g_k e^{-(E_k/K_B T)}}{U_s(T)} \quad (1)$$

where λ is the wavelength of the transition; N_s is the number density (particle/cm³) of the emitting atoms for each species; A_{ki} is the transition probability for the given line; g_k is the k level degeneracy; K_B is the Boltzmann constant; T is the plasma temperature; and $U_s(T)$ is the partition function for the emitting species at the plasma temperature. The emitted intensity is expressed in photons/(s cm³). In actual measurements, the efficiency of the collecting system affects as a scale factor the measured intensity of the line, so that we can write

$$\overline{I}_{ki}^{\lambda} = F C_s A_{ki} \frac{g_k e^{-(E_k/K_B T)}}{U_s(T)} \quad (2)$$

where $\overline{I}_{ki}^{\lambda}$ represents the measured integral line intensity, C_s is the concentration of the emitting atomic species, and F is an experimental parameter that takes into account the optical efficiency of the collection system as well as the plasma density and volume. If some successive runs are needed for covering the whole spectrum or an average over several laser shots is needed in order to enhance the spectral signal, as in our case, it is necessary to take a great care in order to keep fixed the experimental parameters (laser energy, focusing, etc.) during the whole measurement time, for maintaining the parameter F constant.

Description of the Procedure. In Eq. 2, the line integral intensity $\overline{I}_{ki}^{\lambda}$ is the result of the LIPS measurement; the spectroscopic parameters A_{ki} , g_k , and E_k can be retrieved from spectral databases [in our case we got the relevant information from the National Institute for Standards and Technology (NIST) Database for Atomic Spectroscopy]: and the F , C_s , and T values have to be determined from the experimental data. The partition function of each species can be calculated from the spectroscopic data, once the plasma temperature is known.

Let us now define the following quantities:

$$y = \ln \frac{\overline{I}_{ki}^{\lambda}}{g_k A_{ki}}, \quad x = E_k, \quad m = -\frac{1}{K_B T}, \quad (3)$$

$$q_s = \ln \frac{C_s F}{U_s(T)}. \quad (4)$$

Taking the logarithm of Eq. 2 and substituting the above definitions, we obtain the following relationship between the x and y parameters:

$$y = mx + q_s. \quad (5)$$

The two-dimensional space identified by the above-defined x and y coordinates is called the Boltzmann plane. We can write a similar relation for each species in the plasma.

Using the definitions of Eq. 3, we can represent each experimental LIPS line as a point in the Boltzmann plane. Thus, the entire data set of line intensity can be graphically represented as a Boltzmann plot,²⁸ where the different points lie on several parallel straight lines, with slope m and intercept q_s , each one corresponding to a

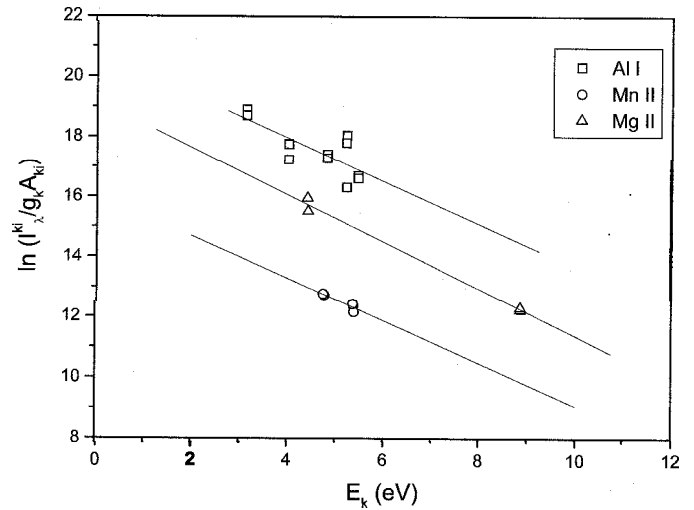


FIG. 1. Boltzmann plot containing some data resulting from the analysis of an aluminum alloy. The three lines represent the results of a linear best fit of the Al(I), Mn(II), Mg(II) data. The slope is the same for the different lines, within the experimental uncertainties.

different atomic species. As an example, a Boltzmann plot obtained from the analysis of an aluminum alloy is reported in Fig. 1.

According to Eq. 4, the slope m of the plots is related to the plasma temperature, while the intercept q_s is proportional to the logarithm of the species concentration.

Let us assume that experimental data are available for each species present in the plasma, that is, in our experimental conditions, for both the neutral and ionized species of each atomic element. In principle, with the hypothesis of LTE, after determination of the plasma temperature, the concentration of a species can be obtained from the measurement of just one experimental line. However, because of the rather large uncertainties existing in the literature on the A_{ki} coefficients, a precise concentration measure must be obtained by using several spectral lines at different energies. Then, the F factor can be determined by normalizing to unit the sum of the species concentration C_s .

$$\sum C_s = \frac{1}{F} \sum U_s(T) e^{q_s} = 1. \quad (6)$$

Finally, the concentration of all the atomic species of the sample can be obtained as

$$C_s = \frac{U_s(T)}{F} e^{q_s} \quad (7)$$

where all the factors are known. The concentration value determined by Eq. 6 refers to one species; in order to obtain the elemental composition, it is sufficient to add the values corresponding to the neutral and single ionized species of the same element; i.e., the total concentration for a given element M is given by

$$C_M^{\text{TOT}} = C_M^{(\text{I})} + C_M^{(\text{II})}. \quad (8)$$

Particular Cases. When only the concentration of one species of a given element is known, it is possible to deduct the concentration of the other ionization stages by making use of the Saha-Boltzmann equation, which re-

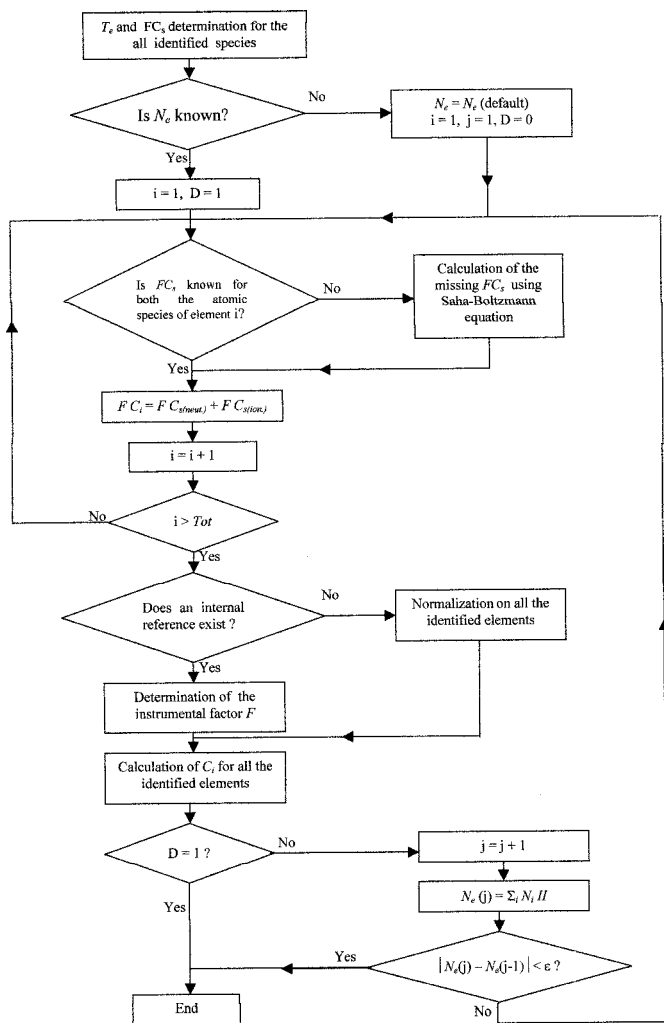


FIG. 2. Flux diagram of the proposed calibration-free algorithm.

lates the concentrations of species of successive ionization stages:

$$\frac{N_e N_{M_0}(\text{II})}{N_{M_0}(\text{I})} = 6.0 \times 10^{21} \frac{g_0^{\text{II}} T^{3/2}}{g_0^{\text{I}}} \exp\left(-\frac{E_{M_0}}{T}\right). \quad (8)$$

where N_e is the electronic density (e/cm^3); $N_{M_0}(\text{I})$ and $N_{M_0}(\text{II})$ are, respectively, the population of the ground state of the neutral atomic species $M(\text{I})$ and that of the single ionized species $M(\text{II})$; g values are the corresponding degeneration values; T is the plasma temperature as obtained by the Boltzmann plots; and E_{M_0} is the ionization potential (in eV) of the species $M(\text{I})$ in its ground state.

Equation 8 enables us to determine the concentration of the ionic species, when the concentration of the other species of the same element is known, if the plasma electron density is known.

This plasma parameter can be obtained by using the same Eq. 8 for an element whose concentration is known for both the ionization states:

$$N_e = 6.0 \times 10^{21} \frac{g_0^{\text{II}} T^{3/2}}{g_0^{\text{I}}} \exp\left(-\frac{E_{M_0}}{T}\right) \frac{N_{M_0}(\text{I})}{N_{M_0}(\text{II})}. \quad (9)$$

Some particularly unfortunate case might occur, in

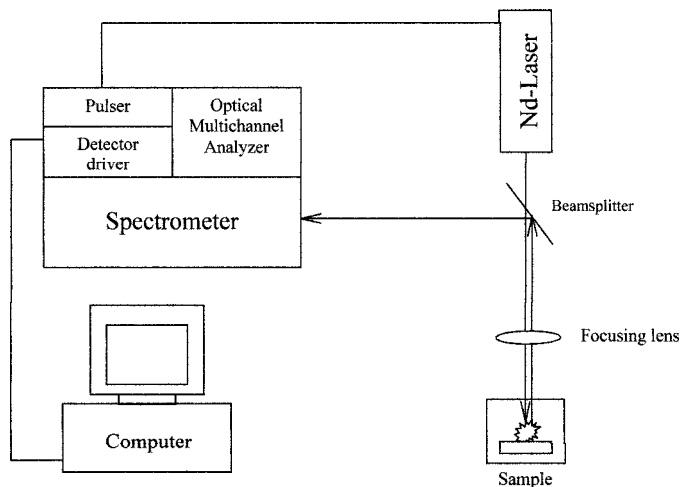


FIG. 3. Schematic view of the LPS experimental apparatus operating at IFAM-CNR Pisa.

which the spectral signal shows lines belonging only to one ionization state for each atomic component. In this case we cannot calculate the N_e value directly by Eq. 9, but an iterative algorithm can be used. First, a reasonable default value of N_e is assumed for the first cycle of calculations; at the end of each cycle N_e is evaluated as the sum of the number of ions resulting from the same calculation (plasma neutrality condition), and substituted to the previous value; the cycle is then repeated up to the convergence of N_e .

The quantitative analysis algorithm above described is sketched in Fig. 2 as a flux diagram that includes the different cases pointed out.

SOME APPLICATIONS OF THE PROCEDURE

Experimental Setup. The experimental apparatus operating in the LIPS laboratory at IFAM-CNR Pisa is shown in Fig. 3. The single parts of the experimental apparatus can be assembled in several ways in order to match different measurement requirements;⁷⁻¹⁵ only the essential components will be briefly described in the following.

The laser source is a Nd:YAG laser, delivering a maximum of 0.4 J in 8 ns at a wavelength of 1064 nm, whose repetition rate can be varied between 1 and 10 Hz. The laser beam is focused on the sample surface by means of a convergent lens of focal length typically ranging from 5 to 30 cm. The optical radiation emitted by the target is collected and focused by using quartz optics (lenses and/or optical fiber) on the input slit of a 1 m spectrometer with a 1200 lines/mm grating. The spectrally dispersed signal is sent to an optical multichannel analyzer (OMA), gated and intensified, which converts it into a digital signal. Finally, the spectral signal is stored and analyzed on a personal computer.

The timing of the whole system is driven by a controller that triggers the laser pulse, starts the OMA acquisition after a suitable delay, and controls the data transfer process from the detector to the computer.

The optimum time delay between the laser pulse and the beginning of the acquisition is chosen in order to maximize the signal-to-noise ratio. Several papers^{10,21}

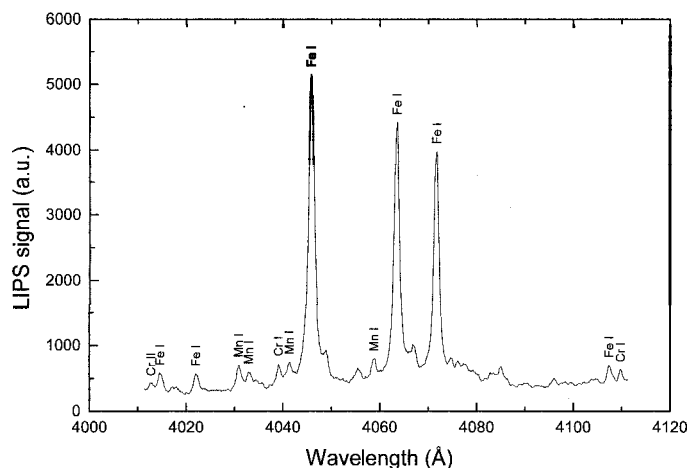


FIG. 4. Portion of spectral signal collected during the analysis of a steel sample. Specific lines emitted by Fe, Mn, and Cr atoms are marked.

have shown the importance of time resolution in LIPS measurements, in order to minimize the effect of continuum recombination radiation on the acquired spectra. In our typical experimental conditions, the choice of a time delay of 1–2 μ s and of an acquisition gate of about 1 μ s allows us to collect a low-background, intense LIPS signal in a regime where space and time variations of plasma parameters are almost negligible.

As already pointed out, in our experimental conditions the plasma spectrum contains specific lines emitted essentially by neutral atomic species and by the species once ionized (the twice ionized species are a negligible fraction of the total population). Figure 4 shows a portion of spectrum collected during the analysis of a steel sample, in our typical experimental conditions.

In the case of solid samples, a translation stage is used for changing the focusing point after a few laser shots, in order to avoid the formation of holes on the sample surface and to preserve the laser-sample interaction geometry. It should be noted that this operation can produce errors in the case of highly inhomogeneous samples, since different sample regions are sampled in the measurement process. However, this problem can be overcome by using new detection systems (echelle spectrom-

TABLE I. Comparison between nominal concentrations and LIPS determined concentrations for an aluminum alloy (anticorrosive).

	LIPS measurement (%)	Nominal concentration (%)
Al	96.9	97.25
Si	1.38	1
Mg	0.65	0.6
Fe	0.46	0.45
Mn	0.33	0.3
Cr	0.13	0.1
Zn	0.074	0.1
Ti	0.045	<0.1
Cu	0.025	<0.05

eters) that allow the acquisition of a whole spectrum with a single laser shot. The use of a stable Nd laser and the precise focusing of the laser beam on the sample are critical conditions for precise quantitative LIPS measurements. Moreover, a configuration where the plasma radiation is collected along the same laser beam axis should be preferred, because of the smaller sensitivity to the fluctuations in spark position occurrence.

The detection devices (optic fiber, monochromator grating and diode array) show a different efficiency for different wavelength ranges. The response curve of the detection system is a characteristic of the experimental apparatus and has been radiometrically determined once and for all in order to normalize the spectral intensities in the whole spectral range considered. It is important to note that only the relative response of the system must be determined, since the information on the absolute intensity of the lines is embedded into the constant parameter F (see Eq. 2).

Preliminary Data Processing. The calibration-free procedure requires all the atomic components of a sample to be detected by measuring at least one of their characteristic spectral lines. Therefore, an unknown sample should be investigated in detail by recording the whole spectrum within the sensitivity range of the spectral detector (2000 to 8000 Å, typically). In our experimental conditions,^{7–15} this is done through several subsequent acquisition runs, each one covering a section of about 100 Å of the whole spectrum. The time required for each run ranges from a fraction of second to some seconds, de-

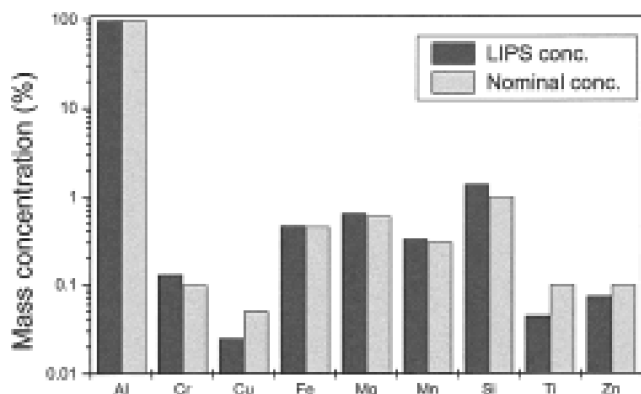


FIG. 5. LIPS quantitative analytical results for an aluminium alloy (anticorrosive), compared with the elemental composition declared by the producer. It must be noted that the nominal values for Cu and Ti are only given by the producer as upper limits.

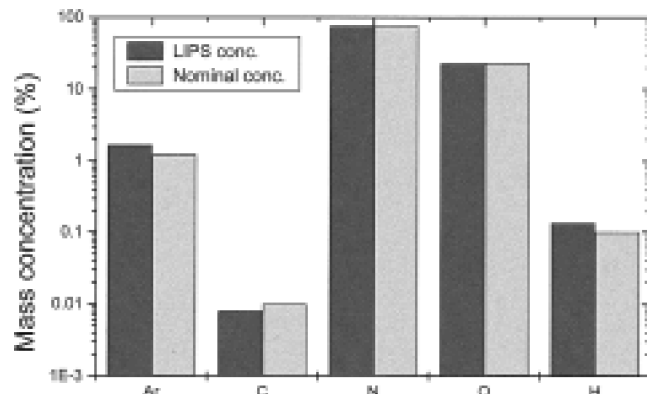


FIG. 6. LIPS quantitative analytical results for atmospheric air, compared with the known composition. For what concerns the hydrogen (from H_2O) and carbon (from CO_2) reference concentration, average values are used corresponding to a typical indoor atmosphere.

pending on the number of pulses used for averaging the LIPS signal. In general, the acquisition of the whole spectrum takes less than 10 min.

The spectral signal collected by the detector is stored on a computer, and the set of data is processed sequentially with the use of a dedicated program developed at IFAM-CNR.

We will describe in the following the general elaboration procedure that applies to all the spectra.

After the raw data have been smoothed by using a Savitzky-Golay algorithm,³⁰ the program performs a search of peaks analyzing the second derivative of the signal intensity vs. the wavelength. The identified lines are then fitted with an analytical approximation of the Voigt function. Among the analytical parameters of the resulting fitting function, the central wavelength and the integral intensity are stored in order to both assign the peak to the corresponding atomic species (qualitative analysis) and determine information about its concentration (quantitative analysis). The integral intensity values are corrected by taking into account the efficiency response of the experimental apparatus at the measured wavelength.

The assignation of a peak to the emitting atomic species is performed by comparison with the set of wavelength data available on the NIST electronic database (NIST DB 38).

Examples of Application of the Calibration-Free Protocol. The procedure here proposed can be applied for the analysis of solid, liquid, and gaseous samples; as examples of its application we report quantitative results obtained for a metallic alloy and for laboratory air.

Figure 5 shows the elemental composition of an aluminum-based alloy (anticorodal), as determined by applying the here described calibration-free LIPS procedure, compared with the nominal composition data given by the producer. The same data are reported in Table I. It is worth mentioning that the LIPS results were obtained in a few minutes, without having recourse to calibration curves. All the elements in the alloy were detected and quantitatively determined in a wide dynamic range, from 250 ppm (Cu) up to 96% (Al).

Figure 6 shows the quantitative analysis of atmospheric air: in this case the LIPS results are compared with the known elemental composition of the atmosphere. Also in this case our calibration-free LIPS measurement protocol gives excellent results over a wide dynamic range.

CONCLUSION

LIPS is a fast and essentially nondestructive diagnostic method, and can be applied to samples of arbitrary shapes.

With the use of the proposed calibration-free procedure, precise quantitative results can be obtained in a few minutes without need of reference samples. Both majority components and trace elements (down to concentrations on order of 1 ppm) can be simultaneously measured.

The application of this new measurement protocol allows one to overcome the so-called "matrix effect", which has been up to now the major obstacle to practical LIPS applications to multielemental quantitative analysis.

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